Computational Mechanics
https://doi.org/10.1007/s00466-025-02619-z

ORIGINAL PAPER

Cut spectral BFS plate element with Lobatto basis for wave propagation analysis

Hela Ambati[1] · Sascha Eisenträger[2] · Santosh Kapuria[1]

Received: 29 October 2024 / Accepted: 2 March 2025
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2025

Abstract
This article presents a cut spectral Bogner-Fox-Schmit (BFS) element for wave propagation analysis of thin isotropic plates of arbitrary shapes at high frequencies. The need for structured discretizations inherent to a recently proposed rectangular spectral BFS element based on C^1-continuous Gauss-Legendre-Lobatto interpolations is addressed by employing a fictitious domain approach, namely the finite cell method. A structured Cartesian mesh is used in place of the conventional geometry-conforming discretization, along with an advanced numerical integration of cut elements. This modification facilitates the analysis of more intricate geometries with non-rectangular shapes or those featuring holes, or inclusions. The free vibration analysis of circular plates demonstrates that this method offers excellent accuracy and convergence properties. Through transverse vibration analysis of a simply supported plate with a hole at the centre, it is shown that responses in low- and high-frequency ranges can be accurately resolved using this method. In this scenario, the numerical results are benchmarked against converged solutions calculated employing the S8R6 element in ABAQUS. The effectiveness of this method in wave propagation problems is verified by comparing its performance with the converged solutions obtained using the standard cut BFS element with a very fine mesh. Results confirm that this element achieves exceptional accuracy, faster convergence, and improved computational efficiency compared to its standard finite element counterpart. The present element, being the first C^1-continuous spectral plate element for general shapes, will facilitate the development of other spectral elements for wave propagation analysis of isotropic and laminated composite plates based on advanced higher-order theories requiring C^1-continuity.

Keywords Spectral BFS element · Cut finite element · Finite cell method · C^1-continuous Lobatto interpolation · Wave propagation analysis

1 Introduction

Significant advancements in using ultrasonic guided waves have revolutionized the approaches within material characterization [1], nondestructive testing (NDT) [2, 3], structural health monitoring (SHM) [4–9], medical diagnosis [10, 11], and communication with quantum systems [12]. This progress prompted the development of novel numerical techniques to enhance computational efficiency across different structural configurations such as beams, plates, and shells [13–15]. This is because most of the above applications deal with high-frequency waves requiring high computational time and also are essentially inverse problems in which both model-based and physics-assisted data-driven approaches require solutions to a large number of wave propagation problems. Moreover, numerical simulations offer a viable solution to minimize the extensive costs associated with experimental research for designing such systems.

Although the classical finite element method (FEM) remains a popular choice for addressing complex challenges, it falls short when it comes to simulating high-frequency guided wave phenomena due to issues like numerical dispersion and dissipation [16–18]. Furthermore, the necessity for

✉ Santosh Kapuria
kapuria@am.iitd.ac.in

Hela Ambati
helaambati.aero.iiests@gmail.com

Sascha Eisenträger
sascha.eisentraeger@ovgu.de

[1] Department of Applied Mechanics, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India

[2] Institute of Mechanics, Otto von Guericke University Magdeburg, 39106 Magdeburg, Germany

Published online: 29 March 2025

dense spatial and temporal discretization to ensure solution convergence imposes a significant computational load [14, 19–21]. Consequently, there has been a concerted effort to develop more sophisticated methods that can accurately and efficiently model wave propagation in structures. Detailed reviews of these advancements are available in [15, 22, 23].

To advance the efficiency of guided wave propagation solutions, a variety of non-conventional and specialized FEM techniques have emerged. Among these, the time-domain spectral element method (SEM), initially introduced by Patera [24], has become especially prominent. Unlike traditional FEM, the SEM leverages specific nodal distributions to derive higher-order orthogonal interpolation functions as shape functions. Rather than opting for a uniform distribution, the SEM places nodes at the roots of orthogonal polynomials, such as Chebyshev and Lobatto polynomials, leading to the spectral rate of convergence enhancing accuracy and efficiency [24, 25]. The SEM has been effectively adapted for one-dimensional (1D) systems like bar structures [26], two-dimensional (2D) domains under plane strain or stress conditions [26], and three-dimensional (3D) solid configurations [27, 28]. The hybrid SEM proposed by Chang and co-workers employs high-order spectral interpolations along the in-plane directions and a linear interpolation for the out-of-plane (thickness) direction for the plate/shell-type structures [29, 30]. This approach is analogous to the ideas proposed by Düster et al. [31] for modelling thin-walled structures via the *p*-version of FEM, which has shown accuracy and computational efficiency. Notably, all the above continuum-based elements require only C^0-continuity for the spectral interpolation of displacements.

As continuum-based elements are generally computationally inefficient for thin-walled structures, spectral elements (SEs) based on structural (beam/plate/shell) theories are essential for wave propagation analysis of such structures. The Reissner-Mindlin plate theory or the first-order shear deformation theory (FSDT) has become a preferred structural theory to develop the time-domain spectral plate elements due to the relative simplicity in achieving C^0-continuity [32, 33]. Similar C^0-continuous SEs have been developed for composite plates integrated with piezoelectric transducers [34, 35]. In contrast, both classical and advanced theories, including the classical plate theory (CPT), the refined third-order theory (TOT) [36, 37], and zigzag theories [38], necessitate C^1-continuity for interpolating deflection, which poses additional complexity in developing conforming finite elements. The latter advanced theories offer highly superior alternatives to the FSDT for incorporating the shear deformation effect for thick isotropic and laminated composite structures and high excitation frequencies as they do not require arbitrary shear correction factors, unlike the FSDT.

One of the simplest conforming C^1-continuous plate elements is the 16-DOF rectangular plate element referred to as Bogner-Fox-Schmit (BFS) element [39], which incorporates the deflection w, slopes $w_{,x}$ and $w_{,y}$ (subscript comma denotes partial differentiation), and, twist $w_{,xy}$ as DOFs at each node. The element uses the tensor product of 1D cubic Hermite interpolation polynomials along two directions and is acclaimed for its remarkable accuracy in handling thin plate problems in rectangular domains, as highlighted by Zienkiewicz et al. [40]. Furthermore, the BFS element, adapted to include complete second-order derivatives [41], has been shown to expedite stress convergence rates. The advantage of the BFS formulation is that it can be easily extended to generate higher-order C^1 continuous plate elements using the tensor product of 1D higher-order generalized Hermite polynomials or its modified version as required by the SEM. However, its applicability to plates having only rectangular sub-domains had limited its use for plates of general shapes. This limitation has been partially mitigated by developing discrete Kirchhoff quadrilateral (DKQ) elements [42–44], in which the deflection and rotation variables are interpolated independently, and the rotation-slope equality of Kirchhoff hypothesis is satisfied at discrete points in the element. While the DKQ concept has been valuable for static and vibration analysis of plates [45], no higher-order version of the element is still available.

Recent advancements have seen the application of isogeometric analysis (IGA) in developing elements for Kirchhoff-plates and Love-shells, which adhere to the C^1-continuity across various shapes [46–48]. Most IGA elements use non-uniform rational B-splines (NURBS) as basis functions, providing high-order smoothness and high geometric flexibility. Like SEM, these elements are also compatible with immersed approaches like the FCM, enabling the simulation of complex geometries without boundary-conforming meshes [49]. Nevertheless, the SEM remains the dominant choice for high-frequency guided wave propagation problems because of its simplicity, robust accuracy, and spectral convergence [15]. In contrast, studies on using IGA for guided wave simulations are scarce. Limited research conducted on C^0-continuous continuum elements suggests that for polynomial orders $p \geq 5$, IGA and SEM produce results of similar quality [14]. However, IGA has some drawbacks compared to SEM. First, its degrees of freedom do not directly correspond to nodal displacements, requiring post-processing to evaluate the displacement field. Second, the stiffness matrix in IGA has a higher condition number than in SEM, which can be problematic when using iterative solvers. Additionally, SEM allows for explicit time integration through mass lumping via under-integration-though this is not yet available for cut elements-whereas IGA currently lacks such a technique. To the best of the authors' knowledge, no study has yet explored IGA-based structural elements for guided wave propagation analysis.

The contributions in developing SFE formulations which employ C^1-continuous interpolation functions include the spectral beam and strip elements based on the classical Euler-Bernoulli theory [50, 51], advanced third-order [36, 37] and zigzag [38] theories, for wave propagation analysis of isotropic and laminated composite structures, and spectral plate element for free vibration analysis of isotropic plates based on the Kirchhoff theory [52]. However, the latter formulation, which employed Chebyshev basis functions for interpolation, resulted in the under-integration of stiffness matrices, introducing significant errors. A thorough assessment of the numerical properties of spectral beam elements was conducted in Ref [53]. The authors [54] have recently presented a C^1-continuous spectral BFS (SBFS) formulation for wave propagation analysis of thin plates, demonstrating its spectral convergence characteristics and high computational efficiency. The formulation utilizes modified generalized bi-Hermite polynomials with Gauss-Legendre-Lobatto (GLL) points as nodal coordinates while incorporating selective collocation for rotational and twisting degrees of freedom (DOFs) at the edges and corners of elements for satisfying C^1-continuity. Despite its promising attributes, the element faces limitations regarding its applicability, confining its use to plates having only rectangular sub-domains.

To broaden the scope of C^1-continuous rectangular plate elements to encompass thin structures of irregular shapes, the fictitious domain approach (FDA) can be applied. Specifically, the finite cell method (FCM) presents a viable solution, as outlined in the works [55, 56]. FCM operates by embedding the original problem geometry within an extended domain, which is then divided into physical and fictitious domains. This division produces discretization that does not conform to the geometry, leading to discontinuous integrands. Consequently, the challenge shifts away from a geometry-conforming discretization towards the accurate computation of system matrices, necessitating a precise evaluation of the discontinuous integrands. The associated issues include the need for advanced numerical techniques to accurately evaluate integrals across elements intersected by boundaries (cut elements), complexities in enforcing essential boundary conditions and addressing conditioning issues related to these cut elements. Proposed solutions to these challenges encompass several quadrature schemes tailored for cut elements, such as the moment fitting technique [57], applying the divergence theorem to reduce problem dimensionality [58], and employing an integration mesh aligning interface elements with element faces [59, 60]. Such approaches effectively bypass traditional meshing problems through sophisticated integration schemes. Various integration methodologies are comprehensively evaluated in Refs [61, 62]. Furthermore, the review papers [63, 64] provide insights into the immersed boundary methods and their practical applications. A mathematical analysis of FCM, demonstrating its exponential convergence, is available in Ref [65].

The first application of the FCM to explicit dynamics of wave propagation was reported in Refs [66–68], where C^0-continuous spectral elements were used in conjunction with the fictitious domain approach with the aim of simplifying the discretization of arbitrary domains. In the context of the classical BFS plate element, Burman et al. [69] presented a cut FCM to extend its applicability for the analysis of thin plates with complex shapes. Eisenträger et al. [70] combined the BFS element with the FCM, echoing the same perspective, and studied its performance for stability analysis. Challenges such as stability and ill-conditioning problems arise, particularly when cut elements contain minimal material [71], prompting the proposal of various stabilization strategies. These include the ghost penalty method [72–74], the fictitious material or α-method [55, 56], and an eigenvalue stabilization technique for nonlinear statics [75] and explicit dynamics [76].

In this work, we combine the spectral BFS element [54] with the FCM [70] for efficient wave propagation analysis of thin plate-type structures of arbitrary shapes. This is an essential step for fast simulation of guided waves in thin-walled isotropic and laminated anisotropic structures based on classical and advanced laminate theories requiring C^1-continuity. The formulation adopts an engineering-centric approach to overcome the stability issues using the indicator function (α-method), where a suitable value is given based on the structural material characteristics. It is acknowledged, however, that more sophisticated solutions, such as customized preconditioning strategies utilizing additive Schwarz methods [71], or the incorporation of supplementary stabilization terms like the frequently employed Ghost penalty in cut FEM [73], while beyond the scope of this paper, represent potential avenues for further exploration and enhancement. The convergence characteristics of the proposed cut spectral BFS (cut SBFS) element are illustrated for natural frequencies of circular plates with respect to its analytical solution. The accuracy and convergence of the element are then assessed for flexural wave propagation in plates with a circular hole. Finally, the element is used to study the interaction of propagating waves with circular and elliptical cut-outs in plates.

2 Time-domain spectral BFS plate element with Lobatto basis

2.1 Weak form of Kirchhoff plate theory

A thin isotropic rectangular plate under transverse loading is considered. The mid-plane of the plate ($z = 0$) is regarded as the reference plane. According to the Kirchhoff plate the-

ory, the variations of the in-plane displacement vector $\mathbf{u}$ and the transverse displacement w at time t along the thickness direction z are approximated as:

$$\boldsymbol{u}(x,y,z,t) = -z\,\boldsymbol{w_{0_d}}(x,y,t),\quad w(x,y,z,t) = w_0(x,y,t),\\ \boldsymbol{u} = \begin{Bmatrix} u_x \\ u_y \end{Bmatrix},\quad \boldsymbol{w_{0_d}} = \begin{Bmatrix} w_{0,x} \\ w_{0,y} \end{Bmatrix}, \tag{1}$$

where $\square_{,x} = \dfrac{\partial \square}{\partial x}$. The weak formulation for the equations of motion of the plate can be expressed in terms of inplane stress vector $\boldsymbol{\sigma} = \begin{bmatrix}\sigma_x & \sigma_y & \tau_{xy}\end{bmatrix}^{\mathrm{T}}$ and inplane strain vector $\boldsymbol{\varepsilon} = \begin{bmatrix}\varepsilon_x & \varepsilon_y & \gamma_{xy}\end{bmatrix}^{\mathrm{T}}$ using Hamilton's principle as

$$\int_A [\langle \rho z^2 \delta \boldsymbol{w_{0_d}}^{\mathrm{T}} \ddot{\boldsymbol{w}}_{\boldsymbol{0_d}} + \rho \delta w_0 \ddot{w}_0 + \delta \boldsymbol{\varepsilon}^{\mathrm{T}} \boldsymbol{\sigma} \rangle - p_z \delta w_0] \mathrm{d}A \\ - \int_{\Gamma_\sigma} \langle -z\sigma_n \delta w_{0,n} - z\tau_{ns}\delta w_{0,s} + \tau_{nz}\delta w_0 \rangle \mathrm{d}\Gamma = 0, \quad \forall\, \delta w_0 \tag{2}$$

where the notation $\langle \square \rangle$ is used to denote integration across thickness, defined as $\langle \square \rangle = \int_{\frac{-h}{2}}^{\frac{h}{2}} (\square)\,\mathrm{d}z$. In Eq. (2), δ denotes first-order variational operator, ρ represents the material mass density, A denotes the surface area of the mid-plane of the plate, $p_z(x,y,t)$ is the normal force per unit area acting on the mid-surface of the plate, Γ_σ denotes the boundary of the plate's mid-plane with n and s as the normal and tangential directions, and σ_n, τ_{ns}, and τ_{nz} are the tractions prescribed on lateral surface at this boundary Γ_σ. On substitution of Eq. (1), Eq. (2) reduces to

$$\int_A \left[\bar{I} \delta \boldsymbol{w_{0_d}}^{\mathrm{T}} \ddot{\boldsymbol{w}}_{\boldsymbol{0_d}} + \hat{I} \delta w_0 \ddot{w}_0 + \delta \boldsymbol{\kappa^0}^{\mathrm{T}} \boldsymbol{M} - p_z \delta w_0 \right] \mathrm{d}A \\ - \int_{\Gamma_\sigma} \left[-M_n \delta w_{0,n} + (M_{ns,s} + V_n)\delta w_0 \right] \mathrm{d}\Gamma \\ + \sum_i \Delta M_{ns}(s_i)\, \delta w_0(s_i) = 0, \quad \forall\, \delta w_0 \tag{3}$$

where s_i denotes the corners in the boundary curve and $\boldsymbol{M} = \begin{bmatrix} M_x & M_y & M_{xy} \end{bmatrix}^{\mathrm{T}}$ are bending stress resultants, $\boldsymbol{V} = \begin{bmatrix} V_x & V_y \end{bmatrix}^{\mathrm{T}}$ are shear stress resultants, and $\boldsymbol{\kappa^0}$ is the curvature vector, defined as

$$\boldsymbol{M} = \langle z\boldsymbol{\sigma} \rangle, \quad \boldsymbol{V} = \langle \boldsymbol{\tau} \rangle, \quad \boldsymbol{\kappa^0} = -\begin{Bmatrix} w_{0,xx} \\ w_{0,yy} \\ 2w_{0,xy} \end{Bmatrix}. \tag{4}$$

The integration process over the thickness also leads to inertia constants $\bar{I} = \langle \rho z^2 \rangle$ and $\hat{I} = \langle \rho \rangle$. In the subsequent development, we will drop the subscript 0 from w_0 for notational simplicity and use only w to mean w_0. Using the linear elastic constitutive relations for an isotropic material $\boldsymbol{M}$ defined in Eq. (4) can be expressed in terms of the curvature κ_0 as

$$\boldsymbol{M} = \mathbf{D}\boldsymbol{\kappa_0}, \qquad \mathbf{D} = \frac{Eh^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}. \tag{5}$$

2.2 Spectral BFS element

Consider a $n \times n$-noded spectral plate element of dimension $l_x \times l_y$ and thickness h having nodal coordinates (ξ_i, η_j) in the natural coordinate system (ξ, η) defined on the interval [-1,1]. The nodes of the proposed spectral plate element are placed at the roots of the GLL polynomials [77] in both directions ξ and η. The nodal coordinates are thus taken as

$$\begin{cases} \xi_1 = -1, \quad \xi_i = r_{i-1}, \quad \xi_n = 1, \\ \eta_1 = -1, \quad \eta_j = r_{j-1}, \quad \eta_n = 1, \quad i,j = 2,\ldots,n-2 \end{cases} \tag{6}$$

where $r_1, r_2, \ldots r_{n-2}$ denote the zeros of the orthogonal Lobatto polynomials, $L_{n-2}(\xi)$ and $L_{n-2}(\eta)$, of $(n-2)^{\text{th}}$ order. The Lobatto polynomial $L_n(\xi)$ of order n is given as [77]

$$L_n(\xi) = \frac{1}{2^{n+1}(n+1)!} \frac{\mathrm{d}^{n+2}}{\mathrm{d}\xi^{n+2}} (\xi^2 - 1)^{n+1}, \qquad \forall\, n \geq 0. \tag{7}$$

The roots of the Lobatto polynomial are obtained using Newton–Raphson method discussed in Refs [50, 54].

The selective collocation of DOFs onto the element nodes, as shown in Fig. 1, is performed to obtain a conforming element while reducing computation time [52, 54]. At the corner nodes, four DOFs are defined, namely the deflection w, the rotations $\theta_x (= w_{,x})$, $\theta_y (= w_{,y})$, and the twist $\theta_{xy} (= w_{,xy})$ as per the classical four-node BFS element. Along the edges normal to the x-direction, at the mid-side nodes, w and θ_x are specified, while w and θ_y are assigned along the edges normal to the y-axis. Only deflection w is defined for the nodes inside the element. The following expresses the relationship between global and local coordinates:

$$x = x_0 + \frac{l_x}{2}\xi, \quad y = y_0 + \frac{l_y}{2}\eta, \quad \frac{\mathrm{d}\square}{\mathrm{d}x} = \frac{2}{l_x}\frac{\mathrm{d}\square}{\mathrm{d}\xi} \\ \text{and} \quad \frac{\mathrm{d}\square}{\mathrm{d}y} = \frac{2}{l_y}\frac{\mathrm{d}\square}{\mathrm{d}\eta}, \tag{8}$$

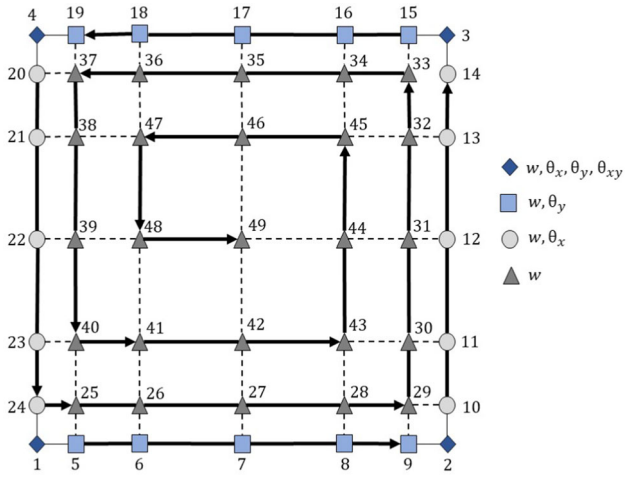


Fig. 1 Node distribution, numbering, and DOFs collocation of a 7×7-noded spectral element

where $x_0 = \dfrac{x_1 + x_n}{2}$ and $y_0 = \dfrac{y_1 + y_n}{2}$. The element displacement vector $\mathbf{U}^e$ is formed as

$$\mathbf{U}^e = \begin{bmatrix} \boldsymbol{u}^e_{\mathbf{11}} & \boldsymbol{u}^e_{\mathbf{21}} \cdots & \boldsymbol{u}^e_{\boldsymbol{ij}} \cdots & \boldsymbol{u}^e_{\boldsymbol{nn}} \end{bmatrix}^{\mathrm{T}}, \tag{9}$$

where

$$\boldsymbol{u}^e_{\boldsymbol{ij}} = \begin{cases} [w_{ij} \quad \theta_{x_{ij}} \quad \theta_{y_{ij}} \quad \theta_{xy_{ij}}]^{\mathrm{T}}, & i, j = 1, n \\ [w_{ij} \quad \theta_{x_{ij}}]^{\mathrm{T}}, & i = 1, n;\ j = 2, \ldots, n-1 \\ [w_{ij} \quad \theta_{y_{ij}}]^{\mathrm{T}}, & i = 2, \ldots, n-1;\ j = 1, n \\ [w_{ij}]^{\mathrm{T}}, & \text{otherwise.} \end{cases} \tag{10}$$

Using Eq. (8), rotations θ_x, θ_y and twist θ_{xy} can be expressed as derivatives with respect to local coordinates:

$$\begin{aligned} \theta_x &= w_{,x} = \frac{2}{l_x} w_{,\xi}\,, \quad \theta_y = w_{,y} = \frac{2}{l_y} w_{,\eta}\,, \\ \theta_{xy} &= w_{,xy} = \frac{4}{l_x l_y} w_{,\xi\eta}\,. \end{aligned} \tag{11}$$

The modified Hermite polynomials for interpolating w using the DOFs shown in Fig. 1 can be derived as [54]

$$\begin{aligned} H_{0i}(\xi) &= \begin{cases} (a_i \xi^2 + b_i \xi + c_i)\, l_i(\xi), & i = 1, n, \\ \dfrac{1 - \xi^2}{1 - \xi_i^2} l_i(\xi), & i = 2, \ldots, n-1, \end{cases} \\ H_{1i}(\xi) &= -\frac{(1 - \xi^2)\xi_i}{2} l_i(\xi), \qquad i = 1, n, \end{aligned} \tag{12}$$

with

$$a_i = \frac{-2\xi_i l_i'(\xi_i) - 1}{4}, \quad b_i = \frac{1}{2}\xi_i, \quad \text{and}$$

$$c_i = \frac{2\xi_i l_i'(\xi_i) + 3}{4}, \qquad i = 1, n. \tag{13}$$

where $H_{ki}(s)$ denotes the Hermite polynomial associated with the kth derivative of the variable w for i^{th} node at ξ_i. The deflection w can be interpolated within the element by employing the tensor product of the 1D polynomials given by Eq. (12) in two directions ξ and η as

$$\begin{aligned} w(\xi, \eta) = \sum_{i=1}^{n} \sum_{j=1}^{n} [&N_{ij}(\xi, \eta)\, w_{ij} + N^x_{ij}(\xi, \eta)\, \theta_{x_{ij}} \\ &+ N^y_{ij}(\xi, \eta)\, \theta_{y_{ij}} + N^{xy}_{ij}(\xi, \eta)\, \theta_{xy_{ij}}]\,, \end{aligned} \tag{14}$$

where N_{ij}, N^x_{ij}, N^y_{ij}, and N^{xy}_{ij} are the C^1-continuous shape functions associated with the DOFs w_{ij}, $\theta_{x_{ij}}$, $\theta_{y_{ij}}$, and $\theta_{xy_{ij}}$, respectively for node (i, j) at (ξ_i, η_j), given by

$$\begin{aligned} N_{ij}(\xi, \eta) &= H_{0i}(\xi) H_{0j}(\eta), && i, j = 1, \ldots, n, \\ N^x_{ij}(\xi, \eta) &= \frac{l_x}{2} H_{1i}(\xi) H_{0j}(\eta), && i = 1, n;\ j = 1, \ldots, n, \\ N^y_{ij}(\xi, \eta) &= \frac{l_y}{2} H_{0i}(\xi) H_{1j}(\eta), && i = 1, \ldots, n;\ j = 1, n, \\ N^{xy}_{ij}(\xi, \eta) &= \frac{l_x l_y}{4} H_{1i}(\xi) H_{1j}(\eta), && i, j = 1, n. \end{aligned} \tag{15}$$

Note that for a four-node element, $n = 2$ for which Eq. (12) leads to the standard cubic Hermite polynomials and Eq. (15) gives rise to the interpolation functions of the BFS element [39]. The displacement w and the curvature vector $\boldsymbol{\kappa_0}$ can be expressed within the rectangular element as

$$w = \mathbf{N}\mathbf{U}^e, \quad \delta w = \mathbf{N}\,\delta\mathbf{U}^e, \quad \boldsymbol{\kappa_0} = \mathbf{B}\mathbf{U}^e, \quad \delta\boldsymbol{\kappa_0} = \mathbf{B}\,\delta\mathbf{U}^e. \tag{16}$$

where $\mathbf{N} = \begin{bmatrix} \boldsymbol{N_{11}} & \boldsymbol{N_{21}} & \ldots & \boldsymbol{N_{ij}} & \ldots & \boldsymbol{N_{nn}} \end{bmatrix}$ and

$$\boldsymbol{N_{ij}} = \begin{cases} [N_{ij} \quad N^x_{ij} \quad N^y_{ij} \quad N^{xy}_{ij}], & i, j = 1, n, \\ [N_{ij} \quad N^x_{ij}], & i = 1, n;\ j = 2, \ldots, n-1, \\ [N_{ij} \quad N^y_{ij}], & i = 2, \ldots, n-1;\ j = 1, n, \\ [N_{ij}], & \text{otherwise.} \end{cases} \tag{17}$$

Fig. 1 illustrates the sequence of node numbering scheme for a 7×7-node element implemented in the solver. The elemental strain–displacement matrix $\mathbf{B}$ in Eq. (16) is given as

$$\mathbf{B} = -\begin{bmatrix} \dfrac{\partial^2}{\partial x^2} \\ \dfrac{\partial^2}{\partial y^2} \\ 2\dfrac{\partial^2}{\partial x \partial y} \end{bmatrix} \mathbf{N} = -\begin{bmatrix} \dfrac{4}{l_x^2}\dfrac{\partial^2}{\partial \xi^2} \\ \dfrac{4}{l_y^2}\dfrac{\partial^2}{\partial \eta^2} \\ \dfrac{8}{l_x l_y}\dfrac{\partial^2}{\partial \xi \partial \eta} \end{bmatrix} \mathbf{N} \tag{18}$$

Substituting Eqs. (16) and (5), the area integrand T shown in the variational Eq. (3) for the entire domain is obtained by summing up the contributions of n_e elements which discretize the domain, as

$$T = \sum_{e=1}^{n_e} \int_{-1}^{1} \int_{-1}^{1} \left[\delta \mathbf{U}^{e\mathrm{T}} \hat{\imath} \mathbf{N}^{\mathrm{T}} \mathbf{N} \ddot{\mathbf{U}}^{e} + \delta \mathbf{U}^{e\mathrm{T}} \mathbf{B}^{\mathrm{T}} \mathbf{D} \mathbf{B} \mathbf{U}^{e} - \delta \mathbf{U}^{e\mathrm{T}} \mathbf{N}^{\mathrm{T}} q \right] \frac{l_x l_y}{4} \mathrm{d}\xi \mathrm{d}\eta$$
$$= \sum_{e=1}^{n_e} \delta \mathbf{U}^{e\mathrm{T}} \left[\mathbf{M}^{e} \ddot{\mathbf{U}}^{e} + \mathbf{K}^{e} \mathbf{U}^{e} - \mathbf{F}^{e} \right]. \tag{19}$$

Summing of element contributions is achieved by a process of assembly of the element matrices $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$ into the corresponding global matrices $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{F}$. This process yields the system equation of motion as

$$\mathbf{M}\,\ddot{\mathbf{U}} + \mathbf{K}\,\mathbf{U} = \mathbf{F}, \tag{20}$$

where $\mathbf{U}$ represents the global vector of DOFs (deflections, rotations, and twists).

3 Finite cell method for cut SBFS element

The proposed SBFS element requires a regular rectangular mesh topology to achieve C^1-continuity between adjacent elements, significantly limiting its applicability to complex geometries. To circumvent this problem, the SBFS element is combined with the FCM to develop the cut SBFS element along the lines presented in Refs [69, 70]. Ensuring C^1-continuity across element boundaries necessitates specific constraints on mesh topology due to the transformation between local and global coordinate systems see Eq. (8). For a smooth transition of the local coordinate axes (ξ, η) between elements, four elements must share a common node. This requirement is inherently met within the current combined approach, which considers regular finite element meshes exclusively. This allows the exceptional accuracy of SBFS elements to be harnessed while accommodating arbitrarily complex geometries without user intervention. A similar FCM implementation for 4-node BFS elements by Eisenträger et al. [70] serves as a foundation.

In FCM, the original physical domain Ω_{P} is replaced by an extended domain Ω_{E} for numerical analysis. The governing equations are solved in Ω_{E}. Ω_{E} is typically regular and simple, leading to a significantly simplified mesh. The proposed approach is confined to void regions within Ω_{E}. Note that it is necessary to use hierarchical high-order enrichment methods for problems involving multiple materials [78]. The Ω_{E} comprises the Ω_{P} and the fictitious domain Ω_{F}, as illustrated in Fig. 2, when only voids are considered. A comparison of mesh generation in FCM and FEA is presented in Fig. 3.

An indicator function $\alpha(\boldsymbol{x})$, acting as a step function that equals one in Ω_{P} and zero in Ω_{F} is introduced. The indicator function $\alpha(\boldsymbol{x})$ is given by

$$\alpha(\boldsymbol{x}) = \begin{cases} 1.0, & \forall \boldsymbol{x} \in \Omega_{\mathrm{P}}, \\ \alpha_0, & \forall \boldsymbol{x} \in \Omega_{\mathrm{F}}. \end{cases} \tag{21}$$

Here, a small constant α_0 is introduced to prevent possible ill-conditioning of the resulting system matrices [55, 62]. While this introduces a minor modelling error, the value of α_0 is carefully chosen to ensure the error remains negligible for practical purposes. Specifically, a value of $\alpha_0 = 10^{-10}$ has been found to provide sufficient numerical stability while keeping the modelling error insignificant for engineering applications. The original problem must now be computed over the entire extended domain. However, the fictitious domain should not influence the behavior of the original problem. To ensure this, all integrals arising in the weak form of the governing equations are multiplied by the indicator function $\alpha(\boldsymbol{x})$. This function scales the stiffness, mass, and force accordingly, making the material in Ω_{F} substantially more compliant than the material in Ω_{P}.

A polynomial function $\mathcal{P}(\boldsymbol{x})$ which is continuous, is considered over a four-sided element $\Omega^{(e)} \in [-1, 1] \times [-1, 1]$. This element has disjoint Ω_{P} and Ω_{F} sub-domains divided by the boundary $\partial\Omega$. In FCM, the primary goal is to integrate $\mathcal{P}(\boldsymbol{x})$ over Ω_{P}. However, this integration needs to be extended over the whole domain Ω_{E}. To exclude the integral's contribution from Ω_{F}, the indicator function $\alpha(\mathbf{x})$ is introduced. As a result, the integral is evaluated as shown, where the integrand becomes discontinuous because of α.

$$\int_{\Omega_{\mathrm{P}}} \mathcal{P}(\mathbf{x}) \mathrm{d}\Omega = \int_{\Omega_E} \alpha(\mathbf{x}) \mathcal{P}(\mathbf{x}) \mathrm{d}\Omega \tag{22}$$

The integral (22) features a discontinuous integrand for all elements intersected by the physical boundary despite the regular integration domain. These elements, distinguished from standard spectral finite elements, are referred to as cut elements (see Fig. 4) and necessitate a special treatment. A specialized integration approach is required since the standard quadrature rules (e.g., Gaussian, Lobatto) commonly used in FEM leads to highly inaccurate results when employed to discontinuous integrands. A quadtree/octree-based decomposition of the cut elements is often preferred due to its flexibility and accuracy, as shown in Fig. 16. Alternative quadrature schemes for cut elements include moment fitting approach [57], the divergence theorem where the dimensionality is reduced by one [58], and an integration mesh where interface elements and element faces coincide [59, 60]. This study employs quadtree-based quadrature for its simplicity and ease of implementation, ensuring fully resolved integration matrices without sacrificing accu-

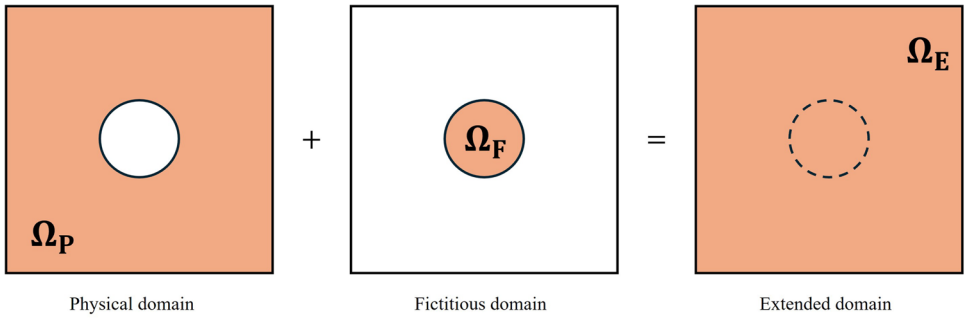


Fig. 2 Fundamental concept of fictitious domain approach

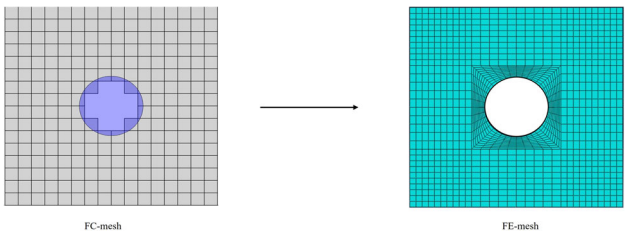


Fig. 3 Comparison of a mesh in FCM with a typical FE mesh

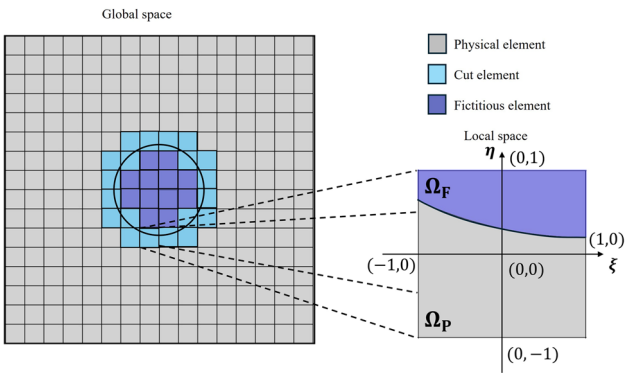


Fig. 4 Fictitious, cut, and physical cells in the local and global spaces

racy, especially in cases with few cut cells. Future research could explore more efficient quadrature techniques, such as moment fitting [57] or reduced integration points for smaller sub-cells. For a broader context, the second author's work discusses various numerical integration schemes, including a merging strategy that reduces integration domains without losing accuracy [79]. Additionally, the Boolean-FCM concept presents another promising approach to enhance computational efficiency [62].

The fundamental concept of the conventional quadtree-based sub-cell integration method is elaborated in Refs [62, 70, 80], which has been implemented here for the cut SBFS elements. The method is briefly recalled in Appendix A. An implicit direct time integration technique, namely the Newmark-β method is used for the semi-discrete equation of motion, as described in Eq. (20). This method provides an unconditionally stable solution [81] and calculates the unknowns for the time step $(t + \Delta t)$ from the previous time step (t) with parameters $\beta = 1/4$, and $\gamma = 1/2$. Currently, no suitable mass lumping approach is available for cut elements. Cut elements should always only be used with a consistent mass matrix; therefore, an efficient time integration demands IMEX-schemes (implicit-explicit time integrators).

4 Numerical results and discussion

In this section, we will start by verifying the cut SBFS element in comparison with analytical solutions for the free vibration response of a free circular plate. We will also assess the achievable rate of convergence for different refinement levels (k_l). Next, we will illustrate the computational advantage of the proposed approach by examining a flexural wave propagation problem in a plate with a circular cut-out, in comparison with the 4-node BFS element combined with FCM [70], referred to as cut BFS element. Finally, we use the element to study the waves's interaction with a circular and an elliptical cut-out in a plate.

4.1 Verification and convergence study

Based on the proposed approach, a `MATLAB` code has been developed for free vibration and transient analysis of plates with cut SBFS elements. To verify the proposed formulation and the computer coding, we obtain the modal responses of a circular plate with free circumference, for which analytical solutions are available [82]. This section examines the convergence rate of the proposed approach for the modal response.

The radius (R) and thickness (h) of the circular plate are 0.5 m and 0.001 m, respectively, and its material properties are $E = 210\,\text{GPa}$ and $\nu = 0.25$, and $\rho = 7800\,\text{kg/m}^3$. For this example, six undamped natural frequencies are considered to obtain the root-mean-square (RMS) error computed as

$$\varepsilon_{\text{RMS}} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(\frac{\omega_i^{\text{num}} - \omega_i^{\text{ref}}}{\omega_i^{\text{ref}}}\right)^2}, \tag{23}$$

where m is the number of modes considered for comparison, ω_i^{num} is the natural frequency of the ith mode determined through the current numerical technique, and ω_i^{ref} is the reference solution which is taken as the analytical solution from Ref [82] derived for this example.

The circular plate is defined using the inequality $(x - x_c)^2 + (y - y_c)^2 > R^2$ in a level set method function, yielding points in the fictitious domain. This function categorizes all cells into three categories as shown in Fig. 4 and determines if a point is inside or outside the physical domain. Here, the parameters are $x_c = 0.75$ m, and $y_c = 0.5$ m.

The following is the derivation of the analytical solution used [82]. The fourth-order governing differential equation of the Kirchhoff plate model in polar coordinates (r, ϕ, t) is given by

$$\rho h \ddot{w} + D\nabla^4 w = 0, \tag{24}$$

where on account of symmetry $\nabla^2 = \partial_{rr} + \frac{1}{r}\partial_r$ and $D = \dfrac{Eh^3}{12(1-\nu^2)}$. The boundary conditions for circular plate with free edge are

$$V_r|_{r=R} = 0 \quad \text{and} \quad M_r|_{r=R} = 0, \tag{25}$$

where V_r and M_r are the transverse shear force and moments per unit length for an edge at r given by

$$\begin{aligned} V_r &= -D\left(\nabla^2 w_{,r}\right) \quad \text{and} \\ M_r &= -D\left[\nabla^2 w - (1-\nu)\frac{1}{r}w_{,r}\right]. \end{aligned} \tag{26}$$

For an eigenvalue problem, we consider $w(r, \phi, t) = W(r, \phi)e^{i\omega t}$, where W is periodic and a function of r. Thus, considering $W = R_m(r)e^{ip\phi}$, Eq. (24) gives

$$\begin{aligned} &\left[\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r} + \left(\gamma^2 - \frac{m^2}{r^2}\right)\right]\left[\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r} - \left(\gamma^2 + \frac{m^2}{r^2}\right)\right] \\ &R_m(r) = 0, \end{aligned} \tag{27}$$

where $\gamma = \dfrac{\omega^2 \rho h}{D}$. The solution $R_m(r)$ for a complete circular plate is given by

$$R_m(r) = C_1 J_m(\gamma r) + C_2 I_m(\gamma r), \tag{28}$$

where $J_m(\gamma r)$ and $I_m(\gamma r)$ are the Bessel function of first kind and modified Bessel function of first kind, respectively. Applying the boundary conditions given in Eq. (25) into Eq. (28), we get the characteristic equation as given below

$$\begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix} = 0, \tag{29}$$

where A_1, A_2, B_1, and B_2 are given by

$$\begin{aligned} A_1 &= m^2(\nu-1)\{(\gamma R)J_m'(\gamma R) - J_m(\gamma R)\} - (\gamma R)^3 J_m'(\gamma R), \\ A_2 &= m^2(\nu-1)\{i(\gamma R)J_m'(i\gamma R) - J_m(i\gamma R)\} + i(\gamma R)^3 J_m'(i\gamma R), \\ B_1 &= (\nu-1)\{(\gamma R)J_m'(\gamma R) - m^2 J_m(\gamma R)\} - (\gamma R)^2 J_m'(\gamma R), \\ B_2 &= (\nu-1)\{(i\gamma R)J_m'(\gamma R) - m^2 J_m(i\gamma R)\} + (\gamma R)^2 J_m(i\gamma R). \end{aligned} \tag{30}$$

When $m = 0$ (no nodal diameter) and $\nu = 0.25$, the above equation simplifies to

$$\frac{J_0(\gamma R)}{J_1(\gamma R)} = \frac{3}{2\gamma R} - \frac{I_0(\gamma R)}{I_1(\gamma R)}. \tag{31}$$

The roots are obtained using inbuilt bessel functions `besselj` (Bessel function of first kind) and `besseli` (modified Bessel function of first kind) available in `MATLAB`. The six natural frequencies that include the first three modes with no nodal diameter ($m = 0$) and the first three modes with one nodal diameter ($m = 1$) are listed in Table 1. The mode shapes for the same obtained using the current method with a very fine mesh are plotted in Fig. 5 using the elements wholly located in the physical domain and discarding all cut elements for visualization.

The convergence behavior of the predicted natural frequencies is studied, taking into account the variation in refinement level k_l is studied. The number of nodes per element edge, n, is varied from 2 to 7 while maintaining a mesh size of 4×4 for the circular plate for different values of k_l, as shown in Fig. 6. As the element order n and refinement level k_l increase, the cut SBFS solution converges to the analytical solution [Eq. (23)], confirming the accuracy of the present formulation. Notably, increasing k_l from 2 to 4 significantly reduces the RMS error when the number of DOFs exceeds 500. However, at lower values of k_l, integration error dominates, causing the error curves to plateau early. Conversely, when the number of DOFs is low and discretization error is high, increasing k_l offers little improvement in accuracy. This highlights the need for a careful balance between numerical integration and discretization errors to achieve optimal accuracy without unnecessary computational costs-a principle well established in the literature on immersed boundary methods.

The convergence plots for the natural frequencies of the circular plate, depicted in Fig. 7 using the proposed method for refinement levels ($k_l = 4$, 5, 6, and 10), showcase the impressive convergence behavior achieved through p-extension (increasing the number of nodes per element edge from $n = 2$ to 9) and h-extension (2×2 and 4×4 mesh). The results show spectral convergence ($\propto (1/n)^n$) under p-extension and exponential convergence with h-refinement, particularly when the integration error tapers off for $k_l > 5$. This demonstrates the effectiveness of the proposed cut SBFS

Table 1 Natural frequencies for six modes of a free circular plate

Nodal diameter m	Mode no.	Analytical solution	ABAQUS soultion (S8R6)	Relative error (%)
0	1	8.75513	8.75511	0.0002
0	2	37.7543	37.7533	0.0026
0	3	86.3165	86.3114	0.0059
1	1	20.0998	20.0983	0.0075
1	2	58.8262	58.8223	0.0065
1	3	117.069	117.058	0.0094
			RMSD (%)	0.0062

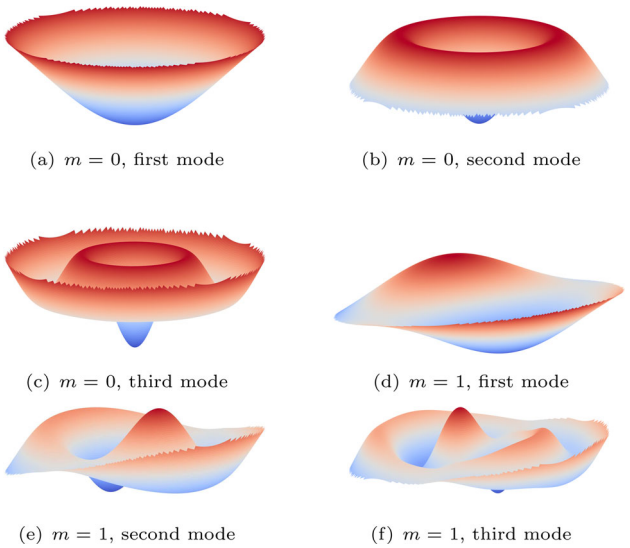


Fig. 5 Six mode shapes for a free circular plate obtained with a very fine mesh of cut SBFS element

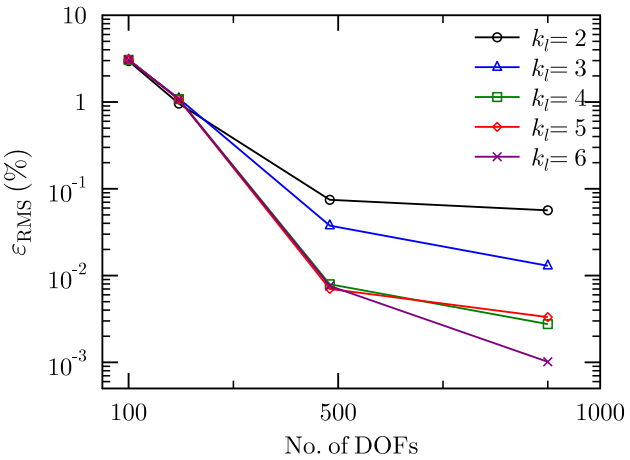


Fig. 6 Convergence of RMS error in six natural frequencies of cut SBFS element model with increasing refinement level k_l for mesh size of 4×4

element, which retains the key characteristics of spectral methods. Furthermore, the steady improvement in convergence with increasing refinement level k_l highlights the power of the FCM employed here.

Additionally, the relative error of individual frequencies and the combined RMS error of the six frequencies obtained employing 7500 S8R6 elements (eight-node shell element with reduced integration) in ABAQUS are compared with respect to analytical solutions and listed in Table 1. For the very fine mesh, ABAQUS model yields an accurate solution with an error less than 0.005%

After verifying the accuracy of natural frequencies for a free circular plate, an analysis of a square plate with a central hole is performed using the proposed approach. The plate has geometric parameters of $L_x = L_y = 1\,\mathrm{m}$, thickness $h = 0.001\,\mathrm{m}$, and a hole with a diameter $D = 0.3\,\mathrm{m}$. The material properties are $E = 210\,\mathrm{GPa}$, $\nu = 0.3$, and $\rho = 7800\,\mathrm{kg/m^3}$. The plate is considered simply supported ($w = 0$) along all outer edges while the hole boundaries remain free. The circular cut-out is defined using the inequality $(x - x_c)^2 + (y - y_c)^2 - R^2 < 0$ in a level set method function to give points in the fictitious domain. The parameters for the cut-out are $R = 0.3\,\mathrm{m}$, $x_c = 0.5\,\mathrm{m}$, and $y_c = 0.5\,\mathrm{m}$. The reference solution is obtained by solving the same problem with a boundary-conforming fine mesh in ABAQUS using 1,200 S8R6 elements.

For the free vibration analysis, we consider the first nine undamped natural frequencies to calculate the RMS using Eq. (23) with $m = 9$ and taking the fine mesh ABAQUS solution The first four mode shapes, obtained using cut SBFS formulation for a very fine mesh, are illustrated in Fig. 8, showing the elements wholly located in the physical domain and discarding all cut elements for visualization.

The all-around simply-supported plate is considered to plot frequency spectra using the cut SBFS element for a regular mesh size of 8 × 8. The natural frequencies ω_i^{num} are normalized against the reference solution ω_i^{ref} obtained using ABAQUS, and plotted with respect to scaled mode number (i/N), where N represents the total number of modes (N = 200 in this case).

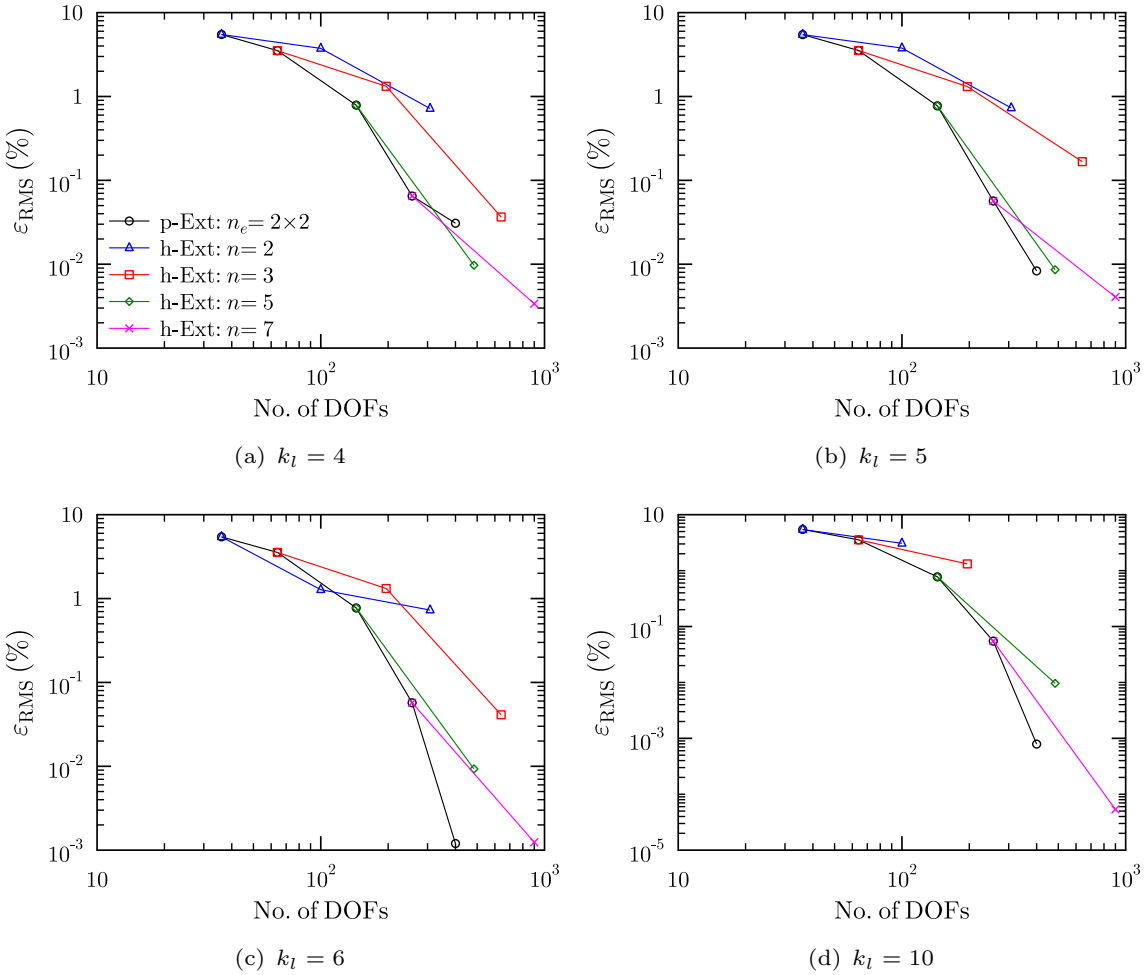


Fig. 7 Convergence behavior in modal analysis of free circular plate for the cut SBFS element with k_l varying from 4 to 10

The present approach provides improved accuracy for lower as well as higher modes with increasing order of the element, with deviations less than 0.5% as shown in Fig. 9a. As the number of nodes per element edge (n) increases, the amplitudes of the oscillatory deviations from the reference solution diminish. As expected, the convergence of the RMS error of the first nine natural frequencies with increasing n is achieved faster for the higher value of the k_l up to $k_l = 5$ (Fig. 9b).

From all the above results, it is evident that the element with $n = 7$ and a refinement level $k_l = 4$ achieves the optimal accuracy with the least number of DOFs. Consequently, we opt for the same for the subsequent wave propagation analysis of a rectangular plate with a hole.

4.2 Wave propagation response of plates with arbitrary cut-outs

In this section, we further assess the efficiency and accuracy of the proposed spectral plate element by investigating wave propagation problems in a rectangular thin plate with circular and elliptical-shaped cut-outs.

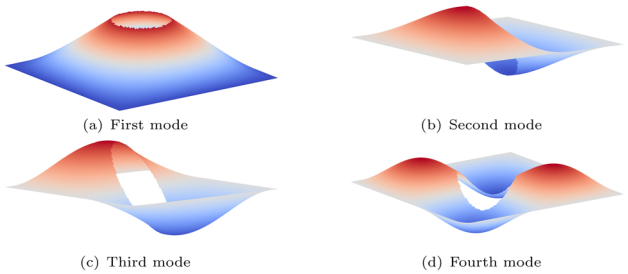
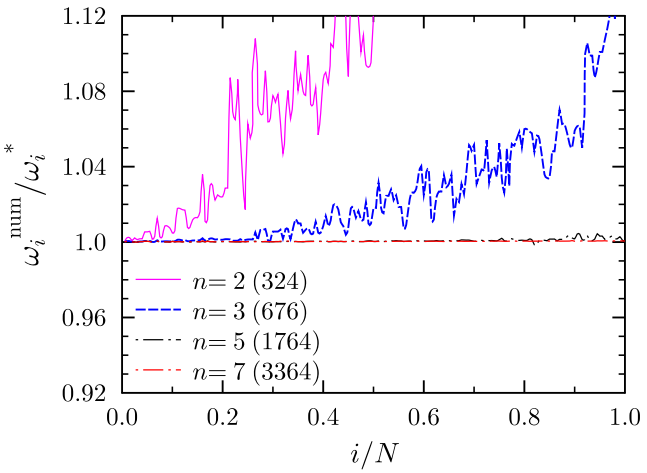
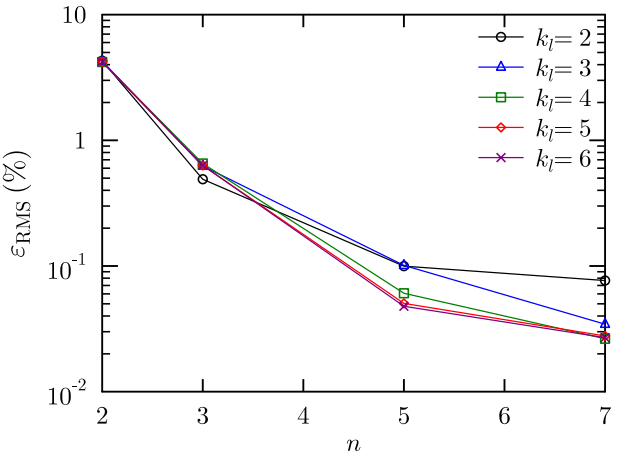


Fig. 8 First four flexural mode shapes for the square plate with a circular hole obtained using a very fine mesh of cut SBFS elements

To evaluate the convergence and accuracy of the developed cut SBFS plate element in transient analyses, we model an aluminum plate with a square-shaped actuator patch of size 7 mm × 7 mm bonded on the top surface. This problem represents real SHM applications where finite-sized patch transducers are used to actuate guided waves. The plate



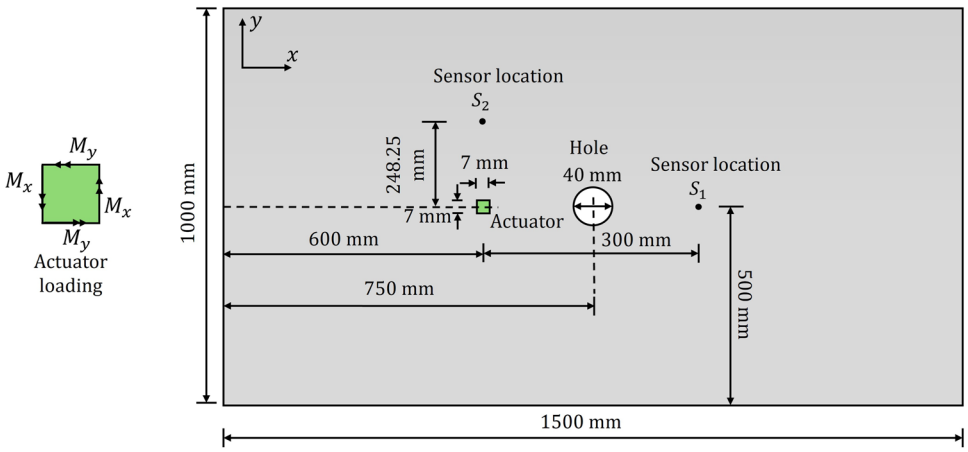
(a) Frequency spectra for the simply-supported plate problem with a hole (p-refinement with a mesh size 8 × 8).



(b) Convergence of RMS error in first nine natural frequencies of cut SBFS element formulation with increasing refinement level k_l from 2 to 6 for mesh size of 4×4.

Fig. 9 Modal analysis for the simply-supported plate problem with a hole

Fig. 10 Configuration of plate with cut-out used for wave propagation analysis



dimensions, with a hole of diameter $D = 40$ mm, are shown in Fig. 10.

The actuation of the patch transducer is simulated using the pin-force method by applying two equal and opposite shear forces $F(t)$ on the two edges of the patch, parallel to the surface [83]. These forces generate distributed moments $M_x(t)$ and $M_y(t)$ that are applied at the actuator edges as shown in Fig. 10, where $M_x(t) = M_y(t) = F(t)h/2$. The moments are applied considering $F(t)$ to be the Hann window-modulated sinusoidal tone burst signal, given by

$$F(t) = \frac{F_0}{2}\left[1 - \cos\left(\frac{2\pi t}{T_H}\right)\right]\sin(2\pi f t), \tag{32}$$

where f represents the pure tone frequency and T_H denotes the width of the pulse, obtained from the ratio of the number of cycles (N_B) in the tone burst signal to the central frequency ($T_H = N_B/f$). In this problem, the structure is excited at 100 kHz frequency with a 5-cycle signal of amplitude $F_0 = 1$ N. The Young's modulus of 70 GPa, mass density ρ of 2700 kg/m^3, and Poisson's ratio ν of 0.3 are considered as the material properties.

A 7 × 7-noded spectral BFS element with $k_l = 4$ for cut elements is employed for the current analysis. The response is monitored at designated sensor locations denoted as S_1 and S_2, positioned at a distance of 300 mm along the $x-$axis and 248.25 mm along the $y-$axis from the actuator. The time-history response of transverse displacements w_1 and w_2 at sensor locations S_1 and S_1, respectively, are computed employing the Newmark-β method with a time step of $\Delta T = 2 \times 10^{-7}$s.

In order to evaluate the accuracy of the current solution, given the absence of an analytical solution, we obtained a reference solution using the standard 2 × 2-node BFS element, which is the lowest order version of the present spectral element combined with FCM [70]. To ensure high accuracy

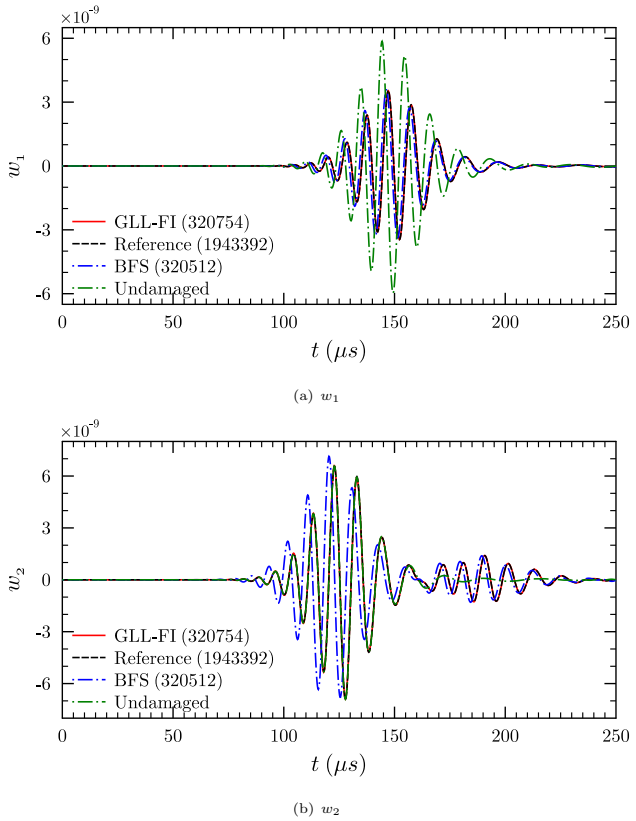


Fig. 11 Transverse displacement-time history at locations S_1 and S_2 of the plate under 5-cycle tone burst excitation for central frequency of 100 kHz

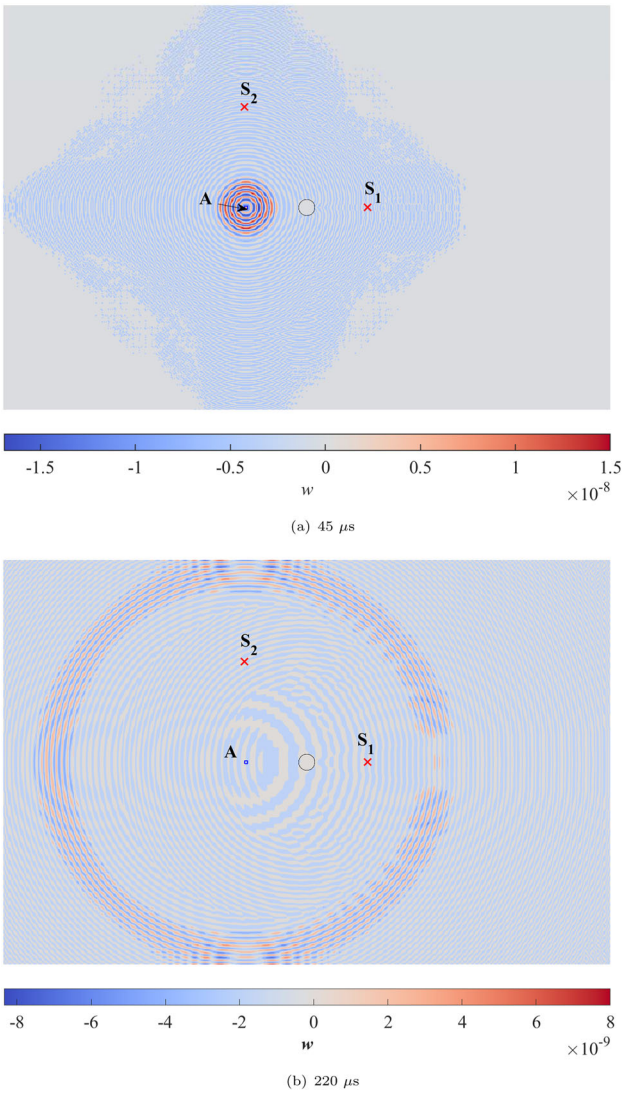


Fig. 12 Contour plots of transverse displacement at time **a** 45 μ s and **b** 220 μs for wave propagation in a thin plate with a circular hole

of the reference solution for the wave propagation problem, it is crucial to adopt a dense mesh network, having at least 20 elements per the minimum wavelength, denoted as $\lambda_{\min}$, according to Ref [20]. $\lambda_{\min}$ is determined from the transverse wave velocity, $c_T = \sqrt{\frac{\mu}{\rho}}$, defined as the minimum phase velocity as $\lambda_{\min} = c_T/f$. When conducting transient analyses, the selected time interval, Δt, needs to be less than the duration for a wave at peak speed, considered here as $c_L = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}}$, to cover the distance l_e between consecutive nodes. Hence, $\Delta t = l_e/c_L$. The adherence to the mesh size guideline $l_e \leq \lambda/20$ simultaneously ensures that the time step is consistent with the requirement for a minimum of 20 steps per wave cycle, or $\Delta t \leq 1/20f$. For all scenarios examined in this analysis, the mesh and time step specifications meet these criteria. Thus, the reference cut BFS formulation is established by considering a very fine mesh comprising 1,943,392 DOFs, employing the Newmark-β scheme with the same values of β and γ as previously stated.

The transverse displacement responses recorded at locations S_1 and S_2 are illustrated in Fig. 11, comparing the current solution obtained with 320,754 DOFs with the reference solution and with that of the pristine plate with no cut-outs [54]. At location S_1, no additional waveform is detected because only the direct wave reaches within the 250 μs duration of the measured response, but its interaction with the hole causes scattering, resulting in a significant reduction in the waveform's amplitude and a delay in its time of arrival (ToA) compared to the undamaged case. At location S_2, an extra waveform is observed due to the wave's reflection from the hole, which does not occur in the undamaged scenario. It is remarkable that the present solution could reproduce the reference solutions, including the interaction with the hole, almost exactly, even though it uses only 1/6 th of DOFs and 1/7 th of computational time in the same computer. In the same figure, we also plot the solutions obtained using the standard cut-BFS element model [70] with an equivalent number (320,512) of DOFs. There is a significant deviation

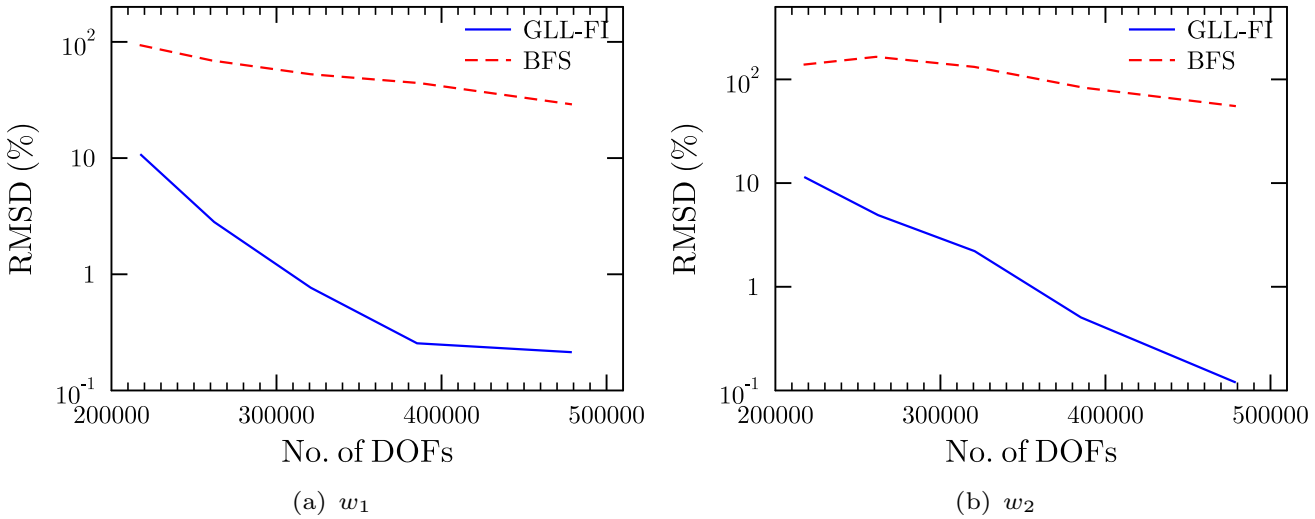


Fig. 13 Convergence of RMSD of the time-history response of w for 100 kHz excitation frequency at sensor locations S_1 and S_2

of its responses from the reference solution in terms of both amplitude and phase velocity, which increases with time as the waves propagate. A contour plot of the deflection field in the plate is shown in Fig. 12 at $t = 45$ μs and 220 μs, indicating the actuator (A) and sensor locations (S_1 and S_2). It is seen that although the actuator shape is square, the wavefront takes a circular shape away from it (see Fig. 12b). At 45 μs, the waveform does not reach the hole, but by 220 μs, it has interacted with the hole, and the resulting extra waveform too has reached the location S_2 in line with what is observed in Fig. 11b.

A quantitative comparative analysis of all the aforementioned solutions is presented in Table 2 in terms of the number of DOFs and percentage root-mean-square deviation (RMSD). The RMSD is calculated using

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^{N_\text{t}} [w_\text{num}(t_i) - w_\text{ref}(t_i)]^2}{\sum_{i=1}^{N_\text{t}} w_\text{ref}^2(t_i)}}, \tag{33}$$

where N_t denotes total number of time steps ($N_\text{t} = T/\Delta t$), $w_\text{num}(t_i)$ and $w_\text{ref}(t_i)$ are the deflection response at i^th time step t_i obtained numerically and from the reference solution, respectively. Furthermore, the convergence of the present and the standard cut BFS element solutions with increasing DOFs is illustrated in Fig. 13. At w_2, where the interaction with a circular hole is dominant, the standard cut BFS element shows much lower convergence than w_1. All the above representations of the results indicate that the present cut SBFS element exhibits much faster convergence and higher accuracy than the standard cut BFS formulation for wave propagation problems in complex structures.

As a last example, the wave propagation analysis of the rectangular plate shown in Fig. 10 with an additional elliptical hole is conducted. The elliptical cut-out used here is defined by implementing the inequality $\frac{(x - x_c)^2}{a^2} + \frac{(y - y_c)^2}{b^2} < 1$ in a level set method function, which returns the information of a point being inside or outside the defined ellipse and is used to categorize all cells into the three categories as shown in Fig. 4. Here, $a = 0.15$ m, $b = 0.075$ m, $x_c = 0.225$ m and $y_c = 0.75$ m.

A contour plot of the transverse displacement response of this plate with both circular and elliptical cut-outs obtained numerically using the proposed approach for 315,283 DOFs is presented in Fig. 14 at 250 μs. The time-history response of w_1 and w_2 at sensor locations S_1 and S_2 are shown in Fig. 14. Comparing this plot with Fig. 11b in which the elliptical cut-out is not present, it is seen the two plots are the same except for the reflections from the elliptical hole appearing in Fig. 15b after the waveform due to interaction with the circular hole. These examples establish that the present cut-SBFS element can be used for faster and more accurate simulations of high-frequency flexural wave propagation in plates of arbitrarily complex shapes.

Table 2 Comparison of the accuracy of the present cut SBFS element with the standard FE for wave propagation problem in a plate with circular hole under 100 kHz modulated tone burst excitation

Sensor location		Reference solution	Present	BFS
S_1	No. of DOF	1,943,392	320,754	320,512
	RMSD (%)	0	0.767	52.7
S_2	No. of DOF	1,943,392	320,754	320,512
	RMSD (%)	0	2.21	132

5 Conclusions

The time-domain cut SBFS plate element is introduced for analyzing flexural guided wave propagation in isotropic plates of non-rectangular geometries at high frequencies. It utilizes a C^1-continuous spectral interpolation based on the modified Hermite polynomials, employing GLL points as the basis and selective collocation of DOFs. The lowest-order version of the element reduces to the classical BFS plate element. The SBFS element is combined with FCM to extend its applicability in thin plates of arbitrary shapes, becoming the first time-domain spectral plate element with C^1-continuity for such non-rectangular plates. Numerical investigations yield the following insights:

1. The combination of the spectral BFS element with FCM provides accurate predictions and optimal convergence rates for free vibration analysis of plates with non-rectangular shapes.
2. The frequency spectra using the proposed cut SBFS element demonstrate improved accuracy for natural frequencies across a large number (200) of modes with increasing order yielding deviations of less than 0.5% for $n \geq 5$.
3. The proposed element showcases exceptional accuracy, faster convergence, and superior computational efficiency compared to the standard cut BFS element for high-frequency wave propagation problems in thin plates with cut-outs of arbitrary shapes.

In future work, the proposed C^1-continuous shape functions will be useful in developing other plate elements based on higher-order theories requiring C^1-continuity for isotropic and composite plates [44, 45, 84]. In the context of fictitious domain approaches, Nitsche's method [85–87] is used to weakly enforce kinematic boundary conditions at curved or inclined edges that do not align with element boundaries, with future research focusing on the specific execution and verification of such approach for the proposed spectral BFS element. For nonlinear problems, the computational efficiency of the implicit integration scheme may be significantly improved by using either the Parallel Direct Solver (PARDISO) [88] or an iterative preconditioned conjugate gradient (PCG) solver [89] with good initial guesses. Additionally, more sophisticated and efficient stabilization techniques can be implemented to enhance the robustness in numerical computations using cut SBFS elements, taking advantage of diagonal mass matrices obtained using such spectral formulations [75, 76]. Comparing the current cut SBFS formulation with immersed IGA will form the scope of a future study.

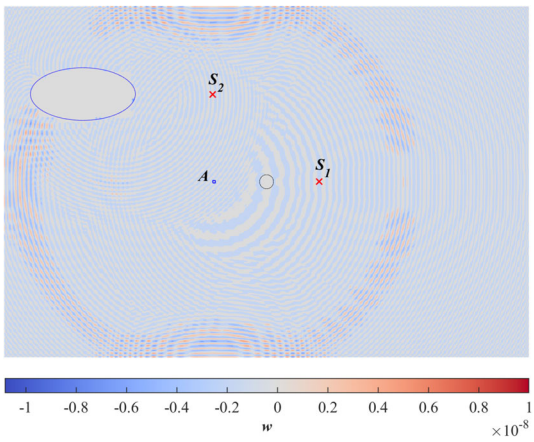


Fig. 14 Contour plots of transverse displacement at time 250 μs for wave propagation problem with both circular and elliptical cut-outs

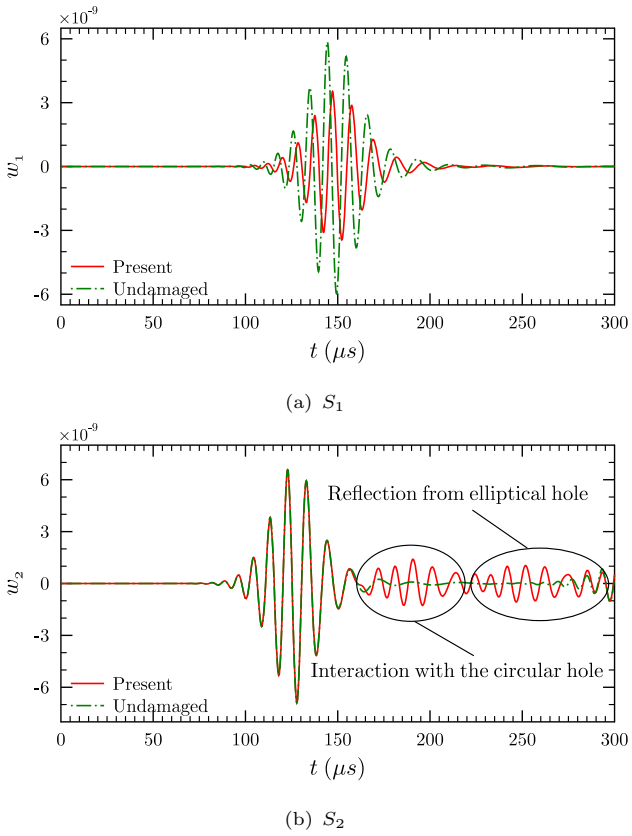


Fig. 15 Deflection-time history at locations S_1 and S_2 of the plate with circular and elliptical cut-outs under 5-cycle tone burst excitation for central frequency of 100 kHz

Appendix A Quadtree integration of cut cells

When dealing with discontinuous integrands, Gaussian quadrature rules prove to be inefficient. Therefore, to minimize the integration error, the domain must be recursively subdivided. This methodology is depicted in Fig. 16 for the cut element shown in Fig. 4. The quadtree decomposition is implemented at various refinement levels k_l. It is evident that the leaf cells (the smallest cells) accumulate around the physical domain's boundary, thereby allowing a composed Gaussian integration rule to produce solutions with better accuracy. Each sub-cell, utilized exclusively for integration, has its local coordinate system (r, s) and is equipped with $(p+1)^2$ integration points, where p is the order of the shape functions ($p = n + 1$), with n being the number of nodes per edge as shown in Fig. 17. Consequently, Eq. (22) is reformulated at the element level as

$$\int_{\Omega_c} \alpha(\mathbf{x})\mathcal{P}(\mathbf{x})\mathrm{d}\Omega = \sum_{i=1}^{n_{\mathrm{sc}}} \sum_{j=1}^{n_{\mathrm{IP}}} \alpha(\mathbf{x}(\xi_{ij}(\mathbf{r}_j)))\mathcal{P}(\mathbf{x}(\xi_{ij}(\mathbf{r}_j)))\mathbf{w}_{ij}, \tag{A1}$$

where $\mathbf{x}$ is the global positioning vector, ξ is the element coordinate system vector, and $\boldsymbol{r}$ is the vector for coordinates within a sub-cell.

In all spectral cut elements, a local integration mesh is introduced as shown in Fig. 16, which subsequently creates smaller and smaller sub-cells near the cut line, which is the discontinuity. Each spectral cut element is divided into four quadrants of equal size and this procedure is repeated until a previously defined refinement level k_l is reached. All sub-cells are again classified as physical, fictitious, and cut. It should be noted that the local integration mesh and the refinement process are introduced for numerical integration only. The total number of DOFs in the system is not increased, as the refinement does not affect the approximation of the primary field variables, which is still defined by the nodal degrees of freedom of the original mesh. The sum of the integrals over the integration sub-cells then yields the integral over the original cut element. The elemental stiffness and mass matrices for cut cells are computed as a summation over the contributions of n_{sc} spectral subcell's stiffness and mass matrices, respectively, given by

$$\mathbf{K}_c^e = \sum_{sc=1}^{n_{sc}} \mathbf{K}_{(c,sc)}^e \quad \text{and} \quad \mathbf{M}_c^e = \sum_{sc=1}^{n_{sc}} \mathbf{M}_{(c,sc)}^e, \tag{A2}$$

where $\mathbf{K}_{(c,sc)}^e$ and $\mathbf{M}_{(c,sc)}^e$ are the stiffness and mass matrices of a specific sub-cell sc in a cut element c is computed in local coordinates r and s, where $(r, s) \in$ [-1,1]. They are given by

$$\begin{aligned}
\mathbf{K}_{(c,sc)}^e &= \int_{-1}^{1}\int_{-1}^{1} \alpha \mathbf{B}^{\mathrm{T}}\mathbf{D}\mathbf{B}\,\det(\mathbf{J}_{sc})\,\det(\mathbf{J})\,\mathrm{d}r\mathrm{d}s, \\
\mathbf{M}_{(c,sc)}^e &= \int_{-1}^{1}\int_{-1}^{1} \alpha\rho h \mathbf{N}^{\mathrm{T}}\mathbf{N}\,\det(\mathbf{J}_{sc})\,\det(\mathbf{J})\,\mathrm{d}r\mathrm{d}s,
\end{aligned} \tag{A3}$$

where $\mathbf{B}$, $\mathbf{D}$ and $\mathbf{N}$ are same as that of given in Sec. 2.2. Due to the usage of the local coordinates r, s of the sub-cells, the matrix $\mathbf{J}_{sc}$ describing the geometry mapping between sub-cell and cut element coordinates which is given by

$$\mathbf{J}_{sc} = \begin{bmatrix} \frac{\partial\xi}{\partial r} & \frac{\partial\eta}{\partial r} \\ \frac{\partial\xi}{\partial s} & \frac{\partial\eta}{\partial s} \end{bmatrix} = \begin{bmatrix} \frac{l_\xi}{2} & 0 \\ 0 & \frac{l_\eta}{2} \end{bmatrix}. \tag{A4}$$

However, for the integration of cut elements in Eq. (A3), the Gauss quadrature rule yields accurate results due to the execution of quadtree decomposition for different k_l values and are obtained as given in Eq. (A5):

$$\begin{aligned}
\mathbf{K}_{c,sc}^e &= \int_{-1}^{1}\int_{-1}^{1} \alpha \mathbf{B}^{\mathrm{T}}(r,s)\mathbf{D}\mathbf{B}(r,s)\det(\mathbf{J}_{sc})\det(\mathbf{J})\,\mathrm{d}r\mathrm{d}s \\
&\approx \sum_{i=1}^{n+2}\sum_{j=1}^{n+2} \alpha W_i W_j \mathbf{B}^{\mathrm{T}}(r_i,s_j)\mathbf{D}\mathbf{B}(r_i,s_j)\,\det(\mathbf{J}_{sc})\,\det(\mathbf{J}), \\
\mathbf{M}_{c,sc}^e &= \int_{-1}^{1}\int_{-1}^{1} \alpha\rho h \mathbf{N}^{\mathrm{T}}(r,s)\mathbf{N}(r,s)\det(\mathbf{J}_{sc})\det(\mathbf{J})\mathrm{d}r\mathrm{d}s \\
&\approx \sum_{i=1}^{n+2}\sum_{j=1}^{n+2} \alpha W_i W_j \rho h \mathbf{N}^{\mathrm{T}}(r_i,s_j)\mathbf{N}(r_i,s_j)\det(\mathbf{J}_{sc})\det(\mathbf{J}).
\end{aligned} \tag{A5}$$

The Gauss points and corresponding weights for a 7 × 7-noded sub-cell is listed in Table 3.

With the aid of quadrature formulae, the element matrices $\mathbf{K}_{c,sc}^e$ and $\mathbf{M}_{c,sc}^e$ are numerically evaluated as shown in Eq. (A5), assuming that E, ρ, and h are constant. If these properties vary, we need to increase the number of integration points depending on the complexity of the distribution of those quantities to obtain accurate results. Further information on the implementation and accuracy of this integration approach can be found in Ref [62].

Acknowledgements S. Kapuria is supported by the Science & Engineering Research Board, Department of Science and Technology, Government of India through the J. C. Bose National Fellowship (Grant No. JBR/2023/000025) and the Core Research Grant (CRG/2021/001237).

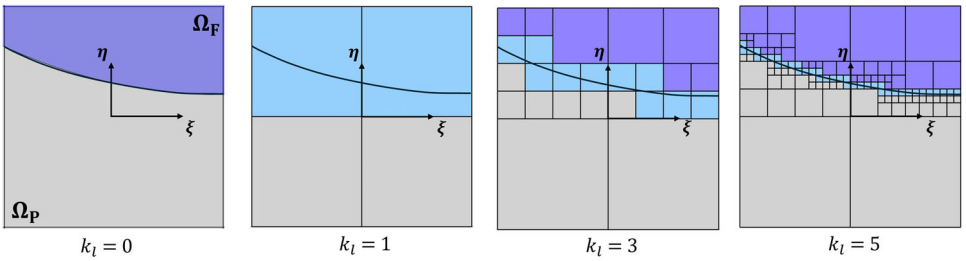


Fig. 16 Different refinement levels $k_l = 0$, 1, 3, and 5 of a cut element in Quadtree decomposition

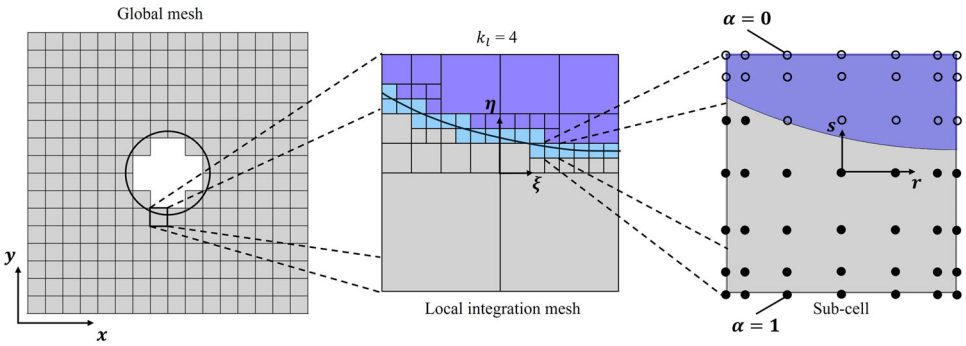


Fig. 17 Coordinate systems considered in the quadtree-based integration

Table 3 Gauss points (r_i / s_j) and corresponding weights (W_i / W_j) for Gauss quadrature

Node no. (i / j)	r_i / s_j	W_i / W_j
1	− 0.9491079123427584	0.1294849661688696
2	− 0.7415311855993945	0.2797053914892768
3	− 0.4058451513773971	0.381830050505119
4	0.0000000000000000	0.4179591836734694
5	0.4058451513773971	0.381830050505119
6	0.7415311855993945	0.2797053914892768
7	0.9491079123427584	0.1294849661688696

References

1. Ponschab M, Kiefer DA, Rupitsch SJ (2019) Simulation-based characterization of mechanical parameters and thickness of homogeneous plates using guided waves. IEEE Trans Ultrason Ferroelectr Freq Control 66(12):1898–1905
2. Rose JL (2014) Ultrasonic guided waves in solid media. Cambridge University Press, Cambridge
3. Wang JJ, Chang TP, Chen BT, Wang H (2012) Determination of Poisson's ratio of solid circular rods by impact-echo method. J Sound Vib 331(5):1059–1067
4. Raghavan A, Cesnik C (2007) Review of guided-wave structural health monitoring. Shock Vibration Digest 39:91–114
5. Agrahari JK, Kapuria S (2016) A refined Lamb wave time-reversal method with enhanced sensitivity for damage detection in isotropic plates. J Intell Mater Syst Struct 27(10):1283–1305
6. Agrahari JK, Kapuria S (2018) Active detection of block mass and notch-type damages in metallic plates using a refined time-reversed Lamb wave technique. Struct Control Health Monit 25(2):e2064
7. Kapuria S, Agrahari JK (2018) Shear-lag solution for excitation, sensing, and time reversal of Lamb waves for structural health monitoring. J Intell Mater Syst Struct 29(4):585–599
8. Kannusamy M, Kapuria S, Sasmal S (2020) Accurate baseline-free damage localization in plates using refined Lamb wave time-reversal method. Smart Mater Struct 29(5):055044
9. Kapuria S (2025) Best reconstruction frequency for time-reversal process of Lamb waves and its determination from a single measurement. Mech Syst Signal Process 223:111887
10. Sarai T, Inoue T, Fujiwara K, Kuramoto K (2012) Dynamic finite element analysis of impulsive stress waves propagating from distal end of femur. Acta Med Okayama 66(5):409–415
11. Li GY, Zheng Y, Jiang YX, Zhang Z, Cao Y (2019) Guided wave elastography of layered soft tissues. Acta Biomater 84:293–304
12. Zhang H, Antoncecchi A, Edward S, Planken P, Witte S (2021) Ultrafast laser-induced guided elastic waves in a freestanding aluminum membrane. Phys Rev B 103(6):064303
13. Gopalakrishnan S (2016) Wave propagation in materials and structures, CRC Press
14. Willberg C, Duczek S, Perez JV, Schmicker D, Gabbert U (2012) Comparison of different higher order finite element schemes for the simulation of Lamb waves. Comput Methods Appl Mech Eng 241:246–261
15. Willberg C, Duczek S, Vivar-Perez JM, Ahmad ZA (2015) Simulation methods for guided wave-based structural health monitoring: a review. Appl Mech Rev 67(1):010803
16. Ihlenburg F, Babuška I (1995) Dispersion analysis and error estimation of Galerkin finite element methods for the Helmholtz equation. Int J Numer Meth Eng 38(22):3745–3774
17. Ihlenburg F, Babuska I, Sauter S (1997) Reliability of finite element methods for the numerical computation of waves. Adv Eng Softw 28(7):417–424

18. Ainsworth M, Wajid HA (2009) Dispersive and dissipative behavior of the spectral element method. SIAM J Numer Anal 47(5):3910–3937
19. Lee R, Cangellaris AC (1992) A study of discretization error in the finite element approximation of wave solutions. IEEE Trans Antennas Propag 40(5):542–549
20. Moser F, Jacobs LJ, Qu J (1999) Modeling elastic wave propagation in waveguides with the finite element method. NDT & E Int 32(4):225–234
21. Han S, Palazotto AN, Leakeas CL (2009) Finite-element analysis of Lamb wave propagation in a thin aluminum plate. J Aerosp Eng 22(2):185–197
22. Palacz M (2018) Spectral methods for modelling of wave propagation in structures in terms of damage detection–a review. Appl Sci 8(7):1124
23. Maio L, Fromme P (2022) On ultrasound propagation in composite laminates: advances in numerical simulation. Prog Aerosp Sci 129:100791
24. Patera AT (1984) A spectral element method for fluid dynamics: Laminar flow in a channel expansion. J Comput Phys 54(3):468–488
25. Rønquist EM, Patera AT (1987) A Legendre spectral element method for the Stefan problem. Int J Numer Meth Eng 24(12):2273–2299
26. Kudela P, Krawczuk M, Ostachowicz W (2007) Wave propagation modelling in 1D structures using spectral finite elements. J Sound Vib 300(1–2):88–100
27. Peng H, Meng G, Li F (2009) Modeling of wave propagation in plate structures using three-dimensional spectral element method for damage detection. J Sound Vib 320:942–954
28. Kim Y, Ha S, Chang FK (2008) Time-domain spectral element method for built-in piezoelectric-actuator-induced Lamb wave propagation analysis. AIAA J 46(3):591–600
29. Ha S, Chang FK (2010) Optimizing a spectral element for modeling PZT-induced Lamb wave propagation in thin plates. Smart Mater Struct 19(1):015015
30. Lonkar K, Chang FK (2014) Modeling of piezo-induced ultrasonic wave propagation in composite structures using layered solid spectral element. Struct Health Monit 13(1):50–67
31. Düster A, Bröker H, Rank E (2001) The p-version of the finite element method for three-dimensional curved thin walled structures. Int J Numer Meth Eng 52(7):673–703
32. Kudela P, Zak A, Krawczuk M, Ostachowicz W (2007) Modelling of wave propagation in composite plates using the time domain spectral element method. J Sound Vib 302(4–5):728–745
33. Kudela P, Ostachowicz WM (2008) Wave propagation modelling in composite plates. Appl Mech Mater 9:89–104
34. Schulte RT, Fritzen CP (2011) Simulation of wave propagation in damped composite structures with piezoelectric coupling. J Theor Appl Mech 49(3):879–903
35. Ge L, Wang X, Wang F (2014) Accurate modeling of PZT-induced Lamb wave propagation in structures by using a novel spectral finite element method. Smart Mater Struct 23:095018
36. Jain M, Kapuria S (2022) Time-domain spectral finite element based on third-order theory for efficient modelling of guided wave propagation in beams and panels. Acta Mech 233(3):1187–1212
37. Jain M, Kapuria S (2022) C^1-continuous time-domain spectral finite element for modeling guided wave propagation in laminated composite strips based on third-order theory. Compos Struct 289:115442
38. Jain M, Kapuria S, Pradyumna S (2022) Efficient time-domain spectral element with zigzag kinematics for multilayered strips. Int J Mech Sci 232:107–603
39. Bogner FK (1965) The generation of interelement compatible stiffness and mass matrices by the use of interpolation formulae, in Proc. Conf. Matrix Methods in Struct. Mech., Airforce Inst. of Tech. Wright Patterson AF Base, Ohio
40. Zienkiewicz OC, Taylor RL, Zhu JZ (2005) The finite element method: its basis and fundamentals. Elsevier
41. Gopalacharyulu S (1973) A higher order conforming, rectangular plate element. Int J Numer Meth Eng 6(2):305–308
42. Batoz JL, Bathe KJ, Ho LW (1980) A study of three-node triangular plate bending elements. Int J Numer Meth Eng 15(12):1771–1812
43. Batoz JL, Tahar MB (1982) Evaluation of a new quadrilateral thin plate bending element. Int J Numer Meth Eng 18(11):1655–1677
44. Kapuria S, Kulkarni S (2007) An improved discrete Kirchhoff quadrilateral element based on third-order zigzag theory for static analysis of composite and sandwich plates. Int J Numer Meth Eng 69(9):1948–1981
45. Kulkarni S, Kapuria S (2007) A new discrete Kirchhoff quadrilateral element based on the third-order theory for composite plates. Comput Mech 39(3):237–246
46. Kiendl J, Bletzinger KU, Linhard J, Wüchner R (2009) Isogeometric shell analysis with Kirchhoff-Love elements. Comput Methods Appl Mech Eng 198(49–52):3902–3914
47. Hughes TJ, Cottrell JA, Bazilevs Y (2005) Isogeometric analysis: CAD, finite elements, NURBS, exact geometry and mesh refinement. Comput Methods Appl Mech Eng 194(39–41):4135–4195
48. Cottrell JA, Reali A, Bazilevs Y, Hughes TJ (2006) Isogeometric analysis of structural vibrations. Comput Methods Appl Mech Eng 195(41–43):5257–5296
49. Schillinger D, Dede L, Scott MA, Evans JA, Borden MJ, Rank E, Hughes TJ (2012) An isogeometric design-through-analysis methodology based on adaptive hierarchical refinement of NURBS, immersed boundary methods, and T-spline CAD surfaces. Comput Methods Appl Mech Eng 249:116–150
50. Kapuria S, Jain M (2021) A C^1-continuous time domain spectral finite element for wave propagation analysis of Euler-Bernoulli beams. Int J Numer Meth Eng 122:2631–2652
51. Han L, Wang J, Li H, Sun G (2020) A time-domain spectral element method with C^1 continuity for static and dynamic analysis of frame structures. Structures 28:604–613
52. Wang J, Li H, Sun G, Han L (2022) Free vibration analysis of rectangular thin plates with corner and inner cutouts using C^1 Chebyshev spectral element method. Thin-Walled Struct 181:110031
53. Eisenträger S, Kapuria S, Jain M, Zhang J (2023) On the numerical properties of high-order spectral (Euler-Bernoulli) beam elements. ZAMM - J Appl Math Mech 103:1–45
54. Ambati H, Eisenträger S, Kapuria S (2024) Time-domain spectral BFS plate element with lobatto basis for wave propagation analysis. Int J Numer Methods Eng p. e7617
55. Parvizian J, Düster A, Rank E (2007) Finite cell method: h-and p-extension for embedded domain problems in solid mechanics. Comput Mech 41(1):121–133
56. Düster A, Rank E, Szabó B (2017) The p-version of the finite element and finite cell methods. Encyclopedia of Computational Mechanics (Second Edition) pp. 1–35
57. Joulaian M, Hubrich S, Düster A (2016) Numerical integration of discontinuities on arbitrary domains based on moment fitting. Comput Mech 57:979–999
58. Duczek S, Gabbert U (2015) Efficient integration method for fictitious domain approaches. Comput Mech 56:725–738
59. Fries TP, Omerović S (2016) Higher-order accurate integration of implicit geometries. Int J Numer Meth Eng 106(5):323–371
60. Kudela L, Zander N, Kollmannsberger S, Rank E (2016) Smart octrees: accurately integrating discontinuous functions in 3D. Comput Methods Appl Mech Eng 306:406–426
61. Abedian A, Parvizian J, Düster A, Khademyzadeh H, Rank E (2013) Performance of different integration schemes in facing discontinuities in the finite cell method. Int J Comput Methods 10(3):1350002

62. Petö M, Duvigneau F, Eisenträger S (2020) Enhanced numerical integration scheme based on image-compression techniques: application to fictitious domain methods. Adv Model Simul Eng Sci 7:1–42
63. Schillinger D, Cai Q, Mundani RP, Rank E (2013) A review of the finite cell method for nonlinear structural analysis of complex cad and image-based geometric models. In: Bader M, Bungartz HJ, Weinzierl T (eds) Advanced computing. Springer, Heidelberg, pp 1–23
64. Schillinger D, Ruess M (2015) The finite cell method: a review in the context of higher-order structural analysis of CAD and image-based geometric models. Arch Comput Methods Eng 22:391–455
65. Dauge M, Düster A, Rank E (2015) Theoretical and numerical investigation of the finite cell method. J Sci Comput 65:1039–1064
66. Duczek S, Joulaian M, Düster A, Gabbert U (2014) Numerical analysis of Lamb waves using the finite and spectral cell methods. Int J Numer Meth Eng 99(1):26–53
67. Joulaian M, Duczek S, Gabbert U, Düster A (2014) Finite and spectral cell method for wave propagation in heterogeneous materials. Comput Mech 54:661–675
68. Duczek S, Liefold S, Gabbert U (2015) The finite and spectral cell methods for smart structure applications: transient analysis. Acta Mech 226(3):845–869
69. Burman E, Hansbo P, Larson MG (2020) Cut Bogner-Fox-Schmit elements for plates. Adv Model Simul Eng Sci 7(1):27
70. Eisenträger S, Kiendl J, Michaloudis G, Duy R, Vetyukov Y (2022) Stability analysis of plates using cut Bogner-Fox-Schmit elements. Comput & Struct 270:106854
71. de Prenter F, Verhoosel CV, van Zwieten GJ, van Brummelen EH (2017) Condition number analysis and preconditioning of the finite cell method. Comput Methods Appl Mech Eng 316:297–327
72. Burman E, Hansbo P (2010) Fictitious domain finite element methods using cut elements: I. A stabilized Lagrange multiplier method. Comput Methods Appl Mech Eng 199(41–44):2680–2686
73. Burman E, Hansbo P (2012) Fictitious domain finite element methods using cut elements: II. A stabilized nitsche method. Appl Numer Math 62(4):328–341
74. Stoter SK, Divi SC, van Brummelen EH, Larson MG, de Prenter F, Verhoosel CV (2023) Critical time-step size analysis and mass scaling by ghost-penalty for immersogeometric explicit dynamics. Comput Methods Appl Mech Eng 412:116074
75. Garhuom W, Usman K, Düster A (2022) An eigenvalue stabilization technique to increase the robustness of the finite cell method for finite strain problems. Comput Mech 69(5):1225–1240
76. Eisenträger S, Radtke L, Garhuom W, Löhnert S, Düster A, Juhre D, Schillinger D (2024) An eigenvalue stabilization technique for immersed boundary finite element methods in explicit dynamics. Comput & Math Appl 166:129–168
77. Pozrikidis C (2005) Introduction to finite and spectral element methods using MATLAB. CRC press
78. Joulaian M, Düster A (2013) Local enrichment of the finite cell method for problems with material interfaces. Comput Mech 52:741–762
79. Petö M, Garhuom W, Duvigneau F, Eisenträger S, Düster A, Juhre D (2022) Octree-based integration scheme with merged sub-cells for the finite cell method: Application to non-linear problems in 3D. Comput Methods Appl Mech Eng 401:115565
80. Duy R (2021) Fictitious domain approach for optimizing stability boundaries of plates with cutouts. Ph.D. thesis, Wien
81. Ostachowicz W, Kudela P, Krawczuk M, Zak A (2011) Guided waves in structures for SHM: the time-domain spectral element method. John Wiley & Sons, Chichester
82. Airey JR (1910) The vibrations of circular plates and their relation to Bessel functions. Proc Phys Soc London 23(1):225
83. Giurgiutiu V (2007) Structural health monitoring: with piezoelectric wafer active sensors. Elsevier, Oxford
84. Kapuria S (2004) A coupled zig-zag third-order theory for piezoelectric hybrid cross-ply plates. J Appl Mech 71(5):604–614
85. Ruess M, Schillinger D, Bazilevs Y, Varduhn V, Rank E (2013) Weakly enforced essential boundary conditions for NURBS-embedded and trimmed NURBS geometries on the basis of the finite cell method. Int J Numer Meth Eng 95(10):811–846
86. Kollmannsberger S, Özcan A, Baiges J, Ruess M, Rank E, Reali A (2015) Parameter-free, weak imposition of Dirichlet boundary conditions and coupling of trimmed and non-conforming patches. Int J Numer Meth Eng 101(9):670–699
87. de Prenter F, Lehrenfeld C, Massing A (2018) A note on the stability parameter in Nitsche's method for unfitted boundary value problems. Comput & Math Appl 75(12):4322–4336
88. Schenk O, Gärtner K, Fichtner W, Stricker A (2001) PARDISO: a high-performance serial and parallel sparse linear solver in semiconductor device simulation. Futur Gener Comput Syst 18(1):69–78
89. Saint-Georges P, Warzée G, Beauwens R, Notay Y (1996) High-performance PCG solvers for FEM structural analysis. Int J Numer Meth Eng 39(8):1313–1340

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.